

application varies for a number of reasons, some are event-driven like launch of a product, and some are time-driven like time of day. Spike testing tests all these issues and guarantees that all the performance-related issues are detected and fixed before the fulfilment of require-

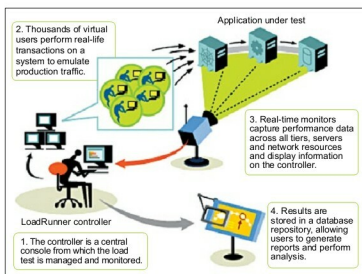


Fig. 3: Working of LoadRunner

The goal of a spike test is to check that when the performance degrades, whether the system will fail or it will handle the load.

Performance testing tools

Performance testing tools include automatic as well as semi-automatic ways of measuring web application performance and load stress capacity. It can monitor the performance of a web application in peak traffic and extreme load conditions. A good performance testing tool must satisfy some criteria such as capability for testing using case-based load scenario testing. It should be easily configurable, capable for stress testing and provide monitoring of parameters.

There are many open source as well as licenced performance testing tools available in the market. A few examples are.

1. HP LoadRunner. HP LoadRunner is an automated performance testing tool and a product from HP for testing the behaviour and performance of systems under actual loads. This tool improves the performance of an application by detecting bottlenecks before a product is launched or deployed. It enables testing of rich Internet applications, Web 2.0 technologies, ERP and CRM applications and legacy applications.

How it works. The HP LoadRunner is a combination of different tools; namely virtual user, generator, control-

ler, load generator and analyser.

The virtual user generator allows, creating scripts that generate a series of actions, such as navigating through the software, amongst others.

The controller takes the script and runs it through a schedule that is set up for testing.

Analyser tells all the results of the load test in various forms. It provides summaries of data, as well as the details of the load test for bottlenecks.

LoadRunner can simulate hundreds or thousands of concurrent users. LoadRunner supports lots of application protocols: Remote Desktop protocol (RDP), Citrix ICA, Flex AF, ERP/CRM, Mail Clients, Database and AJAX Truclient.

Some of the advantages of using LoadRunner are that it is easy to use with nice GUI, does not need installation on servers under testing, uses native monitors, has excellent monitoring and analysis interface and excellent and active support from HP. The only disadvantage associated with this tool is its cost.

2. IBM rational performance tester. Rational performance tester is an automated performance testing tool for web- and server-based applications developed by the Rational Software division of IBM. It is a creation, execution and analysis tool that helps developers to validate the reliability and scalability of a web application before its deployment.

TABLE I
Overview of HP LoadRunner

Version	11.5
Availability	Licensed
Cost	\$50,000 - \$100,000
System requirement	Windows, Linux

TABLE II
Overview of IBM Rational Performance Tester

Version	8.3
Availability	Licensed
Cost	\$1880.00
System requirement	Windows, Linux

How it works. Creation of performance test in IBM Rational performance tester is a three-step process: build scripts, schedule workload, execute and analyse.

Build scripts: Script represents user's activities in the application, scripts are real-world usage scenarios built by the tester based on a design plan—without any programming, and it is built by pointing and clicking through the target application's user interface (UI).

Schedule workload: Schedule defines the sequence and ordering of script execution. It dynamically adds load during the run to find breakpoints.

Execute and analyse: Results after load testing give a clear picture of whether the system can or cannot handle the load. The data provided helps to locate the root cause of a problem.

Protocols supported by Rational Performance tester are Citrix, Socket Recording, SOA, SAP, and Web HTTP.

Advantages:

- (i) Supports XML and is significantly cheaper than LoadRunner
- (ii) No programming knowledge is required
- (iii) Diagnoses Weblogic and Web-sphere application servers
- (iv) Gives you the option of using a persistent cursor for data tables

Disadvantages:

- (i) Does not support Java applet-based application